Spot the Screen Image

Approach: I first tried hand-crafted CV features (FFT frequency peaks, Laplacian variance, color/noise stats, LBP texture, edge density) with an SVM/Random Forest — capped at 82-84%, too sensitive to lighting/angle variation across my dataset (different phones, a laptop, varying brightness). I switched to fine-tuning MobileNetV2 (pretrained on ImageNet): froze most layers, trained only the last 3 blocks + a new classification head, with heavy augmentation (brightness/contrast jitter, rotation, crop, random erasing) to compensate for only 100 training images. Used 320×320 input (above MobileNetV2's default 224) since recapture artifacts like moire and pixel grids are fine-grained and get lost at lower resolution.

Accuracy: 94.0% (± 5.8%) via 5-fold cross-validation on my 100-image dataset. I report the CV mean rather than a single split since 100 images makes any one split noisy. With more data I'd expect this to tighten and likely clear 95%.

Latency: ~45 ms/image on laptop CPU (no GPU at inference) — feels instant.

Cost per image: On-device: ~$0, model is ~9MB and runs locally, no server round-trip.

Cloud alternative: roughly $1 per million images on a small CPU instance (rough estimate).

On-device is clearly preferable here for both cost and privacy.

With more time: mainly more data — doubling/tripling the dataset with more devices, lighting, and objects would most likely close the gap to 95%+ and tighten the CV variance. Secondarily, INT8 quantization (TFLite/ONNX mobile) for a smaller, faster on-device model.